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Characterization of Functional Poly(ethylene oxide)s and Their Corresponding Polystyrene Block Copolymers by Liquid Chromatography under Critical Conditions in Organic Solvents

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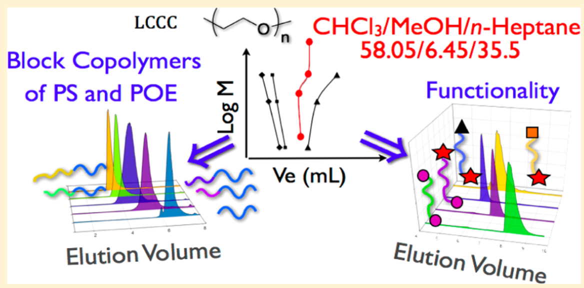
Supporting Information

ABSTRACT: Separation of functional poly(ethylene oxide) PEO and PEO block copolymers was investigated using liquid chromatography under critical conditions (LCCC) with a mixture of organic solvents as eluent. The optimum eluent is a mixture of 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v) using a reverse phase (C_8) column. Unlike what was expected, the elution mechanism is governed by the interaction of a polar end-group with the column. In these conditions, poly(ethylene oxide) (PEO) functionalized with either an acrylate or alkoxyamine moieties were separated. This allows us to investigate the efficiency of the synthesis of poly(ethylene oxide)-*b*-polystyrene (PEO-*b*-PS) and polystyrene-*b*-poly(ethylene oxide) *b*-polystyrene (PS-*b*-PEO-*b*-PS) block copolymers prepared via the combination of 1,2 radical intermolecular addition followed by the nitroxide-mediated polymerization NMP of styrene. Amphiphilic diblock PEO-*b*-PS and triblock PS-*b*-PEO-*b*-PS copolymers were also separated from PEO homopolymers using the same experimental conditions. We showed that the PEO block is then invisible, and the calibration curve obtained using PS homopolymer standards could be used to determine the whole molar mass of the PS block in block copolymers with PS and PEO segments, with a weak influence of the architecture.



INTRODUCTION

Poly(ethylene glycol) (PEG) or poly(ethylene oxide) (PEO) is a hydrophilic polymer which is ubiquitous in numerous biomedical applications owning to its good biocompatibility, such as the lack of toxicity and immunogenicity, non-biodegradability, and ease of excretion from living organisms.[1] Besides its biological properties, the interest of PEO is related to the fact that the polyether backbone is fairly inert to most chemical reactions and soluble in a wide range of organic solvents. In addition, usually PEO has precisely defined terminal functionality, mainly primary hydroxyl groups, which offers the possibility to perform further convenient chemical transformation to enhance and/or tune its chemical reactivity for different applications, e.g., preparation of bioconjugates[2,3] or use as surfactant.[4] These favorable biological and chemical properties, as well as the development of the living/controlled radical polymerization (CLRP) techniques, have stimulated an unprecedented research activity focused on both the chemical derivatization of PEO and the synthesis of PEO-based amphiphilic block copolymers.

In addition, the ability of such block copolymers to self-assemble in ordered nanostructures both in the solid state and in selective solvents opened up the way to a large range of innovative materials. This specific behavior was for instance successfully exploited for the preparation of mesoporous silica with different pore sizes,[5−9] the elaboration of solid polymer electrolytes for rechargeable batteries,[10−14] and the synthesis of metallic nanoparticles.[15−18] Whatever the applications in which PEO materials are used, their performance and properties are strongly influenced by the distribution of the molar mass, the yield of functionality, the chemical composition, the architecture, or the contamination by residual homopolymers.[19,20] Consequently, the quantitative functionalization of PEO and the quality of control are essential in order to obtain precisely defined PEO materials. Moreover, in order to build accurate composition/architecture/properties relationships, a complete and precise characterization of PEO-based materials obtained at different stages of the synthesis process is highly desired.

In this context, one of the most powerful methods allowing for the separation of either different polymers bearing various functional end-groups (telechelic polymers) or block copolymers according to block length is liquid chromatography at critical conditions (LCCC).[21] This chromatographic mode, halfway between size exclusion chromatography (SEC) and interactive liquid chromatography, consists of compensating the entropic and enthalpic interactions between the polymer and the stationary phase. In these conditions, the polymer chain eluted at the critical conditions becomes chromatographically

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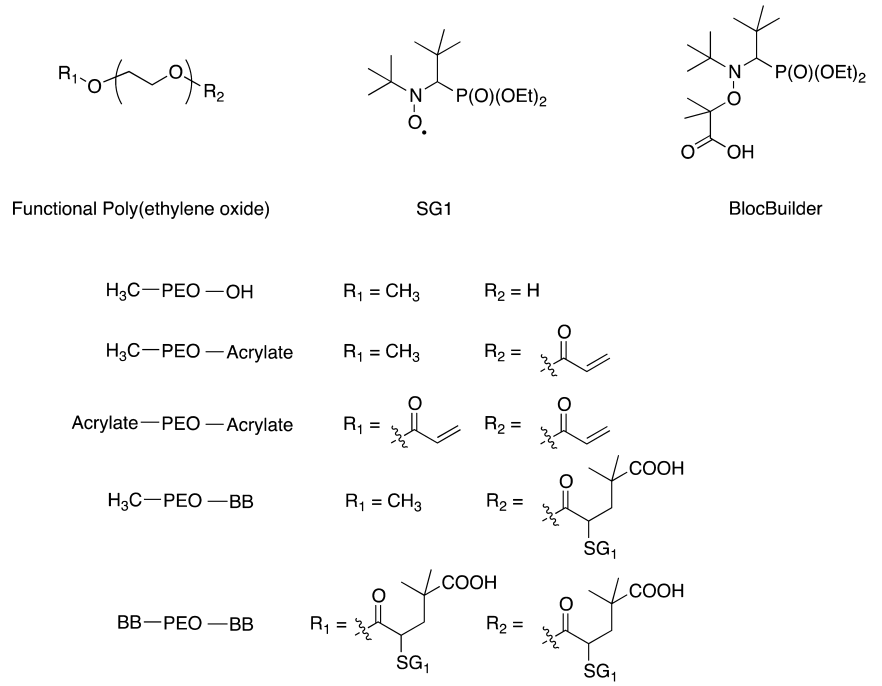
Scheme 1. Structure of Functional Poly(ethylene oxide) (PEO), SG1 Nitroxide, and BlocBuilder Alkoxyamine



"invisible", and the elution is independent of the molar mass. The earliest practical examples of LCCC were focusing on end-group separations.[22] In the past few years, much attention has been paid to the analysis of block copolymers by LCCC. Indeed, information on block length distributions is very difficult to obtain from other methods, such as mass spectrometry or NMR for example.[23,24]

Numerous papers report on the determination of critical conditions for PEO and focused mainly on the separation of diblock and triblock copolymers of PEO and poly(propylene oxide) (PPO).[25−29] Recently, several papers reported the investigations for PEO critical conditions of block copolymers, where the hydrophobic blocks are poly(ε-caprolactone)[30,31] (PCL), poly(L-lactide)[32] (PLA), and polystyrene.[33] In some of these works,[26,29,30] SEC mode elution was promoted for the visible block since it allows for the determination of the length of the block eluted under noncritical conditions and the separation of the critical homopolymer from the copolymer. It has to be noted that in certain cases a block cannot be made completely invisible (especially when the invisible block is long) and that the elution of the whole block copolymer is affected by the length of the block under critical conditions.[34,35] The main point concerning the critical condition for PEO is that all reported studies in the literature were always carried out with a mobile phase composed of water and other organic solvents such as acetone, acetonitrile, methanol, etc. Such mixtures are restricted to PEO-based block copolymers whose second block exhibits a similar solubility[28,29,32] or whose molar mass is sufficiently low[33] to avoid precipitation. However, thanks to the developments of CRP methods and click chemistry approaches, the number of possible composition and architecture based on PEO is constantly increasing. In order to support these important achievements in synthesis, LCCC conditions that allow characterization of PEO-based copolymer exhibiting different polarity for each block and of high molar mass is of high importance but still missing.

In this study, we report original conditions for LCCC of PEO-based materials using only organic solvents as the mobile phase, in which functional PEO (with acrylate and alkoxyamine moieties) and their corresponding block copolymers with polystyrene of different chain architecture: PEO-*b*-PS and PS-*b*-PEO-*b*-PS up to high molar mass (from 10 000 to 70 000 g mol^{-1}) were separated.

EXPERIMENTAL SECTION

Materials. PEO standard (12140 g mol^{-1}) was purchased from Agilent (Polymer Laboratories). PEO of different number-average molar masses (M_n) of 2000, 4600, 9000, 10 000, and 20 000 g mol^{-1} were purchased from Sigma-Aldrich. BlocBuilder MA (>99%) (Scheme 1) alkoxyamine based on the nitroxide SG1 (*N-tert*-butyl-*N*-[1-diethylphosphono-(2,2-dimethylpropyl)] nitroxide) and the 1-carboxy-1-methylethylalkyl moiety was kindly provided by Arkema (France). Acryloyl chloride and triethylamine were purchased from Sigma-Aldrich and were used as received. All solvents and other chemicals were of reagent-grade quality and were obtained commercially and used without further purification.

Block Copolymer Synthesis. The synthetic procedures of CH_3−PEO−acrylate, CH_3−PEO−BB, and CH_3−PEO-*b*-PS as well as that of acrylate−PEO−acrylate, BB−PEO−BB, and PS-*b*-PEO-*b*-PS triblock copolymers have already been described by our group.[36,37] The chemical structure of the different functional PEO is described in Scheme 1.

LCCC Analysis. The LCCC analyses for PEO were performed on a Waters 600 chromatography system equipped with a Waters 600 pump, a Waters 600 controller, and a Waters 717plus autosampler. The following columns were used: Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm. They were placed in a thermostated Crococil oven (Polymer Laboratories), in which a temperature of 30 °C was maintained for all measurements. The mobile phase, a mixture of three organics solvents (58.05 vol % of chloroform, 6.45 vol % of methanol, and 35.50 vol % of *n*-heptane, SDS HPLC grade), was delivered by a Waters 600 pump at a flow rate of 0.8 mL min^{-1}. All solvents were filtered on 0.2 μm PTFE Alltech membrane before use. Samples were prepared at a concentration of 0.25 wt % in the same mixture than that of the mobile phase and were also filtered on a 0.2 μm PTFE Sodipro filter and injected using a Waters 717plus autosampler equipped with a 20 μL loop. A Polymer Laboratories PL-ELS2100 detector was used at an evaporation temperature of 70 °C and a nebulization temperature of 40 °C. Nitrogen was used as a carrier gas at a flow rate of 1.4 mL min^{-1}. It has to be noted that the ELSD detector give a nonlinear response and that detector has not been used to quantify the different species.

NMR Analysis. NMR experiments were performed with a Bruker Avance III 400 MHz Nanobay spectrometer. ^{1}H NMR spectra were recorded at 300 K with an 12.7 μs 30° pulse, a repetition time of 2 s and 128 scans. The ^{1}H NMR spectrum of the final PEO-*b*-PS copolymer in $CDCl_3$ shows peaks (δ = 6.5−7.2 ppm) corresponding to the phenyl protons of the PS block and peaks (δ = 3.2−4.4 ppm) corresponding to the $-CH_2-CH_2-O-$ protons of the PEO block. The final average copolymer composition was thus determined from the ratio between the integrals of PS-phenyl ^{1}H signals and the PEO ^{1}H signals. Considering the molar mass values given by the supplier for the PEO block, the ^{1}H NMR analysis led to the determination of the PS block average molar masses that are given in Table 1.

Table 1. PEO-*b*-PS Block Average Copolymers Composition Determined by ^{1}H NMR

copolymer	$M_{n(PEO)}$ (kg mol^{-1})	$M_{n(PS)}$ (kg mol^{-1})	$M_{n(total)}$ (kg mol^{-1})
diblock A	10.0	58.3	68.3
diblock B	2.0	21.4	23.4
diblock C	10.0	20.0	30.0
diblock D	2.0	11.2	13.2
diblock E	10.0	3.6	13.6
triblock A′	10.0	54.4	64.4
triblock B′	10.0	9.0	19.0
triblock C′	10.0	5.4	15.4

RESULTS AND DISCUSSION

Critical Conditions for PEO. Critical conditions for the PEO were determined using Nucleosil (highly deactivated) C_8 columns and four commercially available PEO standards with different molar masses (4600, 9000, 12 140, and 20 000 g mol^{-1}). These standards are usually synthesized via living anionic polymerization initiated by a difunctional initiator and terminated by the addition of a proton donor such as methanol and are therefore α,ω-dihydroxylated PEO. When the mobile phase was only composed of pure chloroform (a good solvent for the polymer), PEOs were eluted in adsorption mode, whereas we could expect the opposite since a reverse phase column is used. This could be explained by the strong interactions of the polar EO units with the unmodified free silanol groups still present in the stationary phase.[38] The addition of 10 vol % of methanol in the mobile phase suppresses or at least lowers these undesired polymer−column packing interactions and PEOs were eluted in the SEC mode as expected. Indeed, methanol is a strong polar eluent enabling strong silanophilic interactions and is then able to suppress the interaction of the stationary phase with the polar segment of PEO. It has to be noted that the chloroform and methanol mixture (90/10 v/v) is thermodynamically a good solvent for PEO (desorli). *n*-Heptane, as PEO nonsolvent (adsorli), was then added to the preceding mixture to compensate for the entropic exclusion effect and then reach the critical conditions. The logarithm of the PEO molar mass ranging from 600 to 20 000 g mol^{-1} versus the elution volume on the Nucleosil C_8 HD columns was plotted (Figure 1) according to the mobile phase composition (chloroform−methanol−*n*-heptane).

As observed in Figure 1, the retention mechanism changes from SEC to LAC depending the composition of the mobile phase. The mobile phase composition to reach the critical conditions was determined to be 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v). For high molar masses (above 20 000 g mol^{-1}), a S-shaped curvature in the evolution of the molar mass versus the retention volume was observed (Figure 2) as already reported in the literature.[38−41] To explain this phenomenon, the groups of Wang[40] and Berek et al.[38,41] suggested the presence of sites at the pore surface having different interaction energies with the polymer-repeating

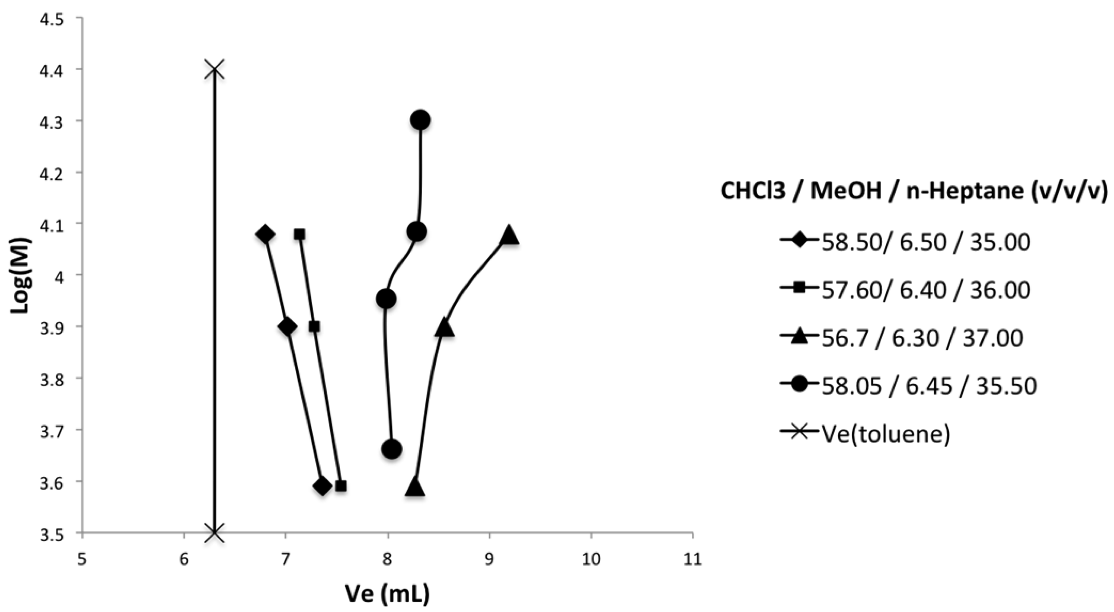


Figure 1. Log (molar mass) vs elution volume (V_e) plot of standard PEO for various mobile phase compositions near the critical conditions. Stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm (at 30 °C), and flow rate was 0.8 mL min^{-1}.

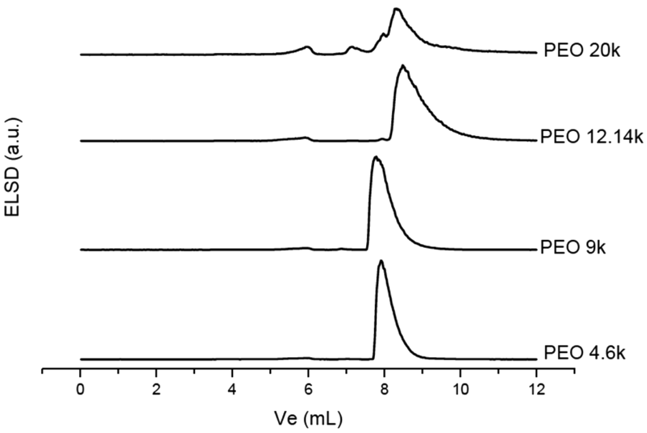


Figure 2. Chromatograms of standard PEO in the critical conditions.

units. Some of these sites are more attractive toward the polymer chain than the others. As the polymer chain increases in length, it could interact with some "minority" stronger attractive surface sites, whereas short polymer chains cannot bridge over these sites and therefore they are being submitted to stable and lower enthalpic attraction.

Besides, a broadening of the elution peaks with increasing molar masses (Figure 2) was also observed. According to the literature,[38−41] this is due to the pore size of the column packing material. Indeed, Gorbunov et al.[39] as well as Cramail et al.[42] have demonstrated that the deviation to the critical conditions occurring in small pore size columns (typically 10 nm) could be overcome by using larger pore size columns. However, the choice of the pores size column must take into consideration a compromise between the SEC mode for the visible block and the critical point for the invisible block. In this work, critical conditions are only reached for a limited range of molar masses (<12 000 g mol^{-1}), which is related to the pore size of the column used.

The vertical line in Figure 1 indicates the retention volume (about 6.3 mL) of toluene injected in the same chromatographic conditions compared to standard PEOs. According to the theoretical model, the chromatographic distribution coefficient K_d equals unity at this retention volume, and polymers under these conditions should elute exactly at the same volume. However, as can be seen in Figure 1, standard PEOs under near critical conditions elute with higher elution volumes, meaning that K_d must be greater than unity. The deviation of the elution point of PEO from the solvent peak position in the reversed phase HPLC has been observed already.[43] In that case, the authors explained this phenomenon by hydrophobic interaction. Indeed, in reverse phase HPLC of PEO, water is always used as one of the eluent, and the entropy gain due to the release of water molecules when PEO chains are adsorbed to the stationary phase is larger than the conformational entropy loss of the polymer chains, leading to a entropy-driven process ($\Delta S^0 > 0$). This results in a distribution coefficient K_d different from unity.

In our case, the occurrence of hydrophobic interaction is less envisioned since the aim of this study was to find LCCC conditions without water. The obtention of a distribution coefficient K_d above unity is probably due to the very polar OH end-groups of PEO chains, which could potentially interact with residual free silanols still present on the column packing material even after the use of methanol to deactivate or at least to hinder them. Obviously, whatever the nature of the interaction, this causes a moderate retention of the polymer chains by the stationary phase. This explanation is partially confirmed by the retention of functional POE after derivatization of the OH group, whose elution is close to the retention volume of toluene (see below for details).

Separation of Functional PEO. PEO bearing various end-groups were synthesized from commercially available mono and dihydroxy-PEOs. Their structures are presented in Scheme 1. The presence of these groups should induce a difference in the interaction with the pore surface, leading to a separation of these polymers in the critical conditions exclusively by the functionality. The critical conditions determined above were then tested for the separation of polymer chains according to their chain-end functionality. According to the results displayed in Figure 3, these conditions allowed good separation of the tested functional PEO. We expected that, using reversed phase columns, the retention volume of PEO functionalized with an alkyl end group (i.e., CH_3−PEO−OH) must be higher than the one with a dihydroxyfunctionalized standard PEO (HO−PEO−OH).[27] However, we noticed that all monofunctional and difunctional PEO analyzed having less polar end-groups were eluted before the standard PEO. Surprisingly, the elution order seemed to be opposite to the one expected for a C_8 reversed phase column. We assigned this phenomenon to an interaction between the polymer end-groups with some residual free silanols even after deactivation by methanol still present on the column packing material as already discussed above.

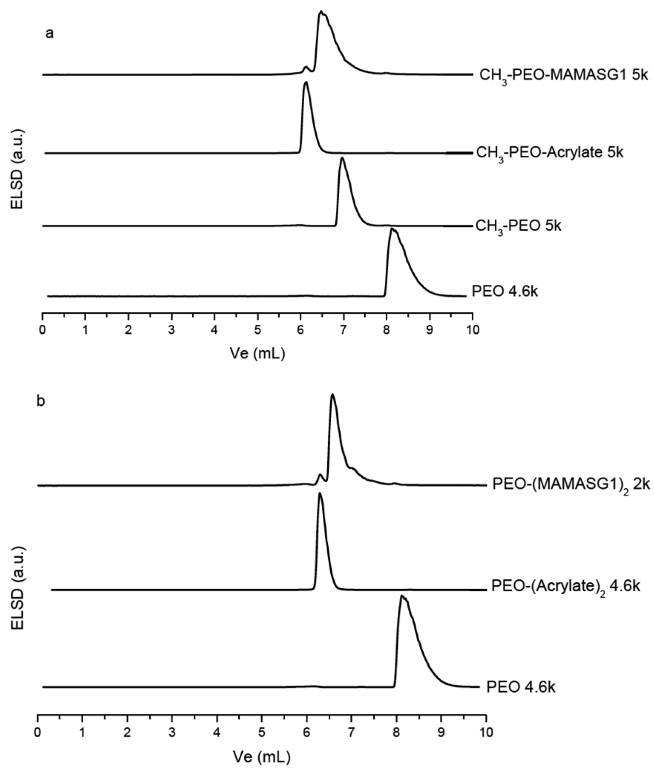


Figure 3. Chromatograms of monofunctional (a) and difunctional (b) PEO analyzed in the critical conditions of standard PEO. Stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm (at 30 °C), mobile phase composition was 58.05% chloroform, 6.45% methanol, and 35.50% n-heptane (v/v/v), and flow rate was 0.8 mL min^{-1}.

According to the obtained results, the polarity of the end-groups increases in the following order: acrylate < BlocBuilder < OH. The polarity of the BlocBuilder end-group results from a combination of the polar carboxylic acid function and the phosphonate group that could lead to strong attractive interactions with the residual free silanols of the stationary phase. In the same time the presence of some apolar methyl and *tert*-butyl groups that have the opposite effect are also present on the polymer chain end. This leads very likely to moderate the polarity of such functional PEO compare to the OH and acrylate end-groups. Even if all functional PEO samples were well separated in our LCCC conditions, the effective interaction parameters for different end-chain groups cannot be determined because the distribution coefficient K_d for PEO standards is greater than unity. It has to be noted that in the case of CH_3−PEO−acrylate the elution volume is close to the solvent elution volume that corresponds to a distribution coefficient K_d close to one.

Next we exploited the critical conditions determined for functional PEO to monitor the preparation of PEO−BB and BB−PEO−BB macroalkoxyamines that are efficient macro-initiators valuable in nitroxide-mediated polymerization. Typically, the synthesis of the mono- or di-PEO macroalkoxy-amine consisted of two steps following previously reported procedures.[36,37] Briefly, the first step is the esterification of either the HO−PEO−Me or HO−PEO−OH with acryloyl chloride to form the corresponding PEO (di)acrylate. The second step corresponds to the 1,2 intermolecular radical addition of the BlocBuilder[44−46] onto the acrylate group to yield the corresponding PEO-based macroalkoxyamines. Obviously, the purity of PEO-based block copolymers is closely related to the functionalization yield for each step of the synthetic procedure. Indeed, a partial functionalization of PEO in the first step leads to a mixture of hydroxyl-terminated PEO and PEO−acrylate, while an uncompleted functionalization in the second step leads to a macroalkoxyamine contaminated by the presence of PEO−acrylate. The chromatograms for PEO before and after each functionalization step are presented in Figure 3a,b. The analyses indicated that the esterification reaction was quantitative since neither the peaks of CH_3−PEO−OH nor that of HO−PEO−OH were observed in the PEO−acrylate chromatograms. Concerning the 1,2 radical intermolecular addition reaction, the analyses shown that the reaction was incomplete. Indeed, the mono- and difunctional PEO-macroalkoxyamine chromatograms present a small peak at 6.2 mL, i.e., the same retention volume to that of the PEO-acrylate obtained in the preceding step, which can be attributed to unreacted PEO−acrylate. It is known that on model-activated olefins (i.e., butyl acrylate) the consumption of the double bond is not complete, and less than 5% of the starting olefin still remains unreacted.[44,46] Nevertheless, in the case of polymer chains (M_n close or above 2000 g mol^{-1}), NMR is not sensitive enough since the ^{1}H spectrum (see Supporting Information for details) did not show any peak corresponding to acrylic protons.

This complementary analysis showed that (i) the proportion of residual PEO−acrylate present in the PEO−macroalkoxy-amine products is very likely negligible, (ii) the LCCC technique is a powerful method to characterize functionalized polymers where other methods failed, and (iii) LCCC is a useful technique to monitor the chemical derivatization of polymer end-groups.

Characterization of POE-*b*-PS and PS-*b*-POE-*b*-PS. Two series of standard PS (PS A: 377.4, 96.0, 19.7, 4.5, and 1.2 kg mol^{-1} and PS B: 188.7, 46.5, 9.9, 2.4, and 0.6 kg mol^{-1}) have been analyzed at the critical conditions of standard PEO (see Figure 4) to establish the calibration curve of the PS block.

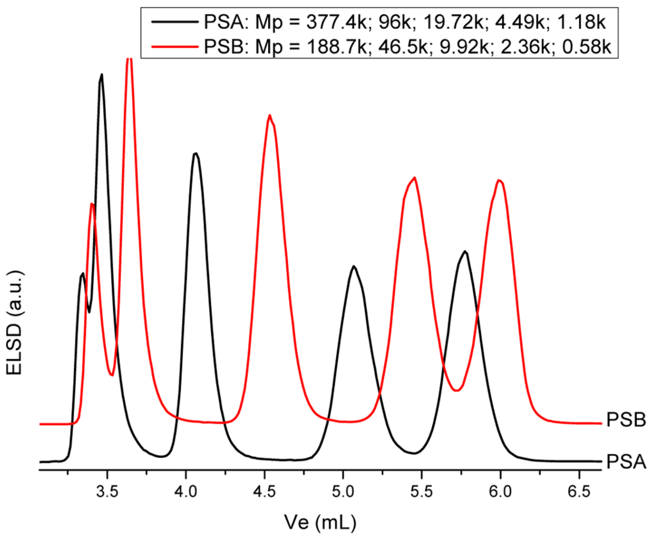


Figure 4. Chromatograms of standard PS analyzed in the critical conditions of standard PEO. Stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm (at 30 °C), mobile phase composition was 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v), and flow rate was 0.8 mL min^{-1}.

Indeed, homopolymers of PS are eluted following the size exclusion mode. As expected, a good resolution is obtained up to a molar mass of 96 kg mol^{-1}. Above this molar mass the separation decreases due to the pore size column packing material. As PEO is chromatographically invisible, this calibration should allow for the determination of the whole PS blocks masses in the PEO-*b*-PS block copolymers.

The calibration curve obtained using these standards is reported in Figure 5. At low molar mass the elution volume is close to the solvent elution volume (K_d = 1) but strongly different from PEO standards due to the interactions of the polar OH groups with the remaining free silanols (see above for details). This result let envision a problem to use the calibration curve for determining the molar mass of the PS block since the "invisibility" of the POE block is hampered by a strong interaction with the column (K_d > 1). Nevertheless, at such volume (6.3 mL) the retentions of low-M_n PS and CH_3−PEO−acrylate are rather similar. This showed that the noninvisibility of the POE block is only due to the hydroxyl end-group, and if the POE is linked to the block copolymer, there is no problem anymore and we could thus expect to determine the PS block length using the calibration curve obtained from PS standards.[47,48]

Two kinds of block copolymers were analyzed at the critical conditions of standard PEO: PEO-*b*-PS (see Figure 6a) and PS-*b*-PEO-*b*-PS (see Figure 6b). A small system peak, also present in the blank chromatogram (not presented here), at about V_e = 8 mL is observed.

As the quantity of PEO−acrylate is below the NMR quantification limit and knowing the molar mass of the PEO block, the average molar mass of PS has been determined using

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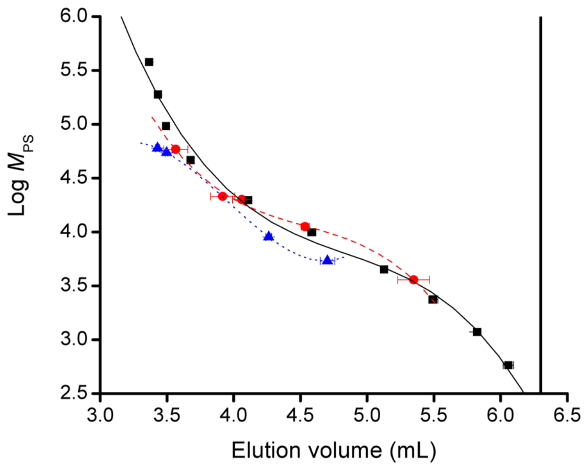

Figure 5. Log $M_{n,PS}$ vs elution volume plot of standard PS (— and ■), diblock copolymers PEO-*b*-PS (● and - - -), and triblock copolymers PS-*b*-PEO-*b*-PS (▲ and ···) in the critical conditions of standard PEO. Vertical line at V_e = 6.3 mL shows the total volume of the columns. Error bars were calculated by calculating the standard deviation (σ) on five injections and then assuming an error of $\pm 2\sigma$. The stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 12 0 Å, particle size of 3 μm (at 30 °C), mobile phase composition was 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v), and flow rate was 0.8 mL min^{-1}.

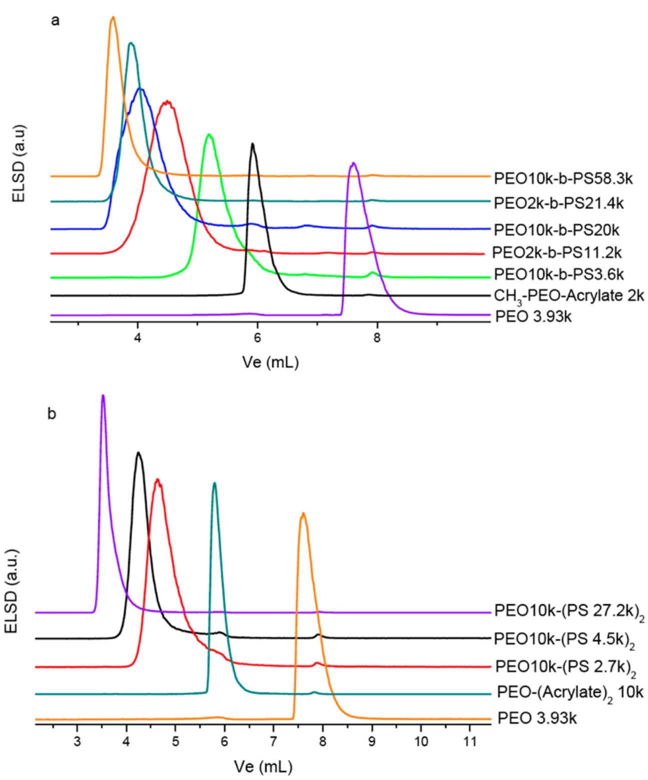

Figure 6. Chromatograms of (a) diblock and (b) triblock copolymers analyzed in the critical conditions of standard PEO. Stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm (at 30 °C), mobile phase composition was 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v), and flow rate was 0.8 mL min^{-1}.

^{1}H NMR by comparison of the characteristic signals of each blocks. Before such calculations, we ensured that the block copolymer did not contain any residual PS homopolymer that could distort the results (see Supporting Information for details).

The peak attributed to block copolymers elutes at a volume ranging between 3 and 6 mL. As the PS homopolymers, the elution of the diblock and triblock followed the SEC mode. For example diblock A PEO(10k)-*b*-PS-(58.3K) is eluted before diblock C PEO(10k)-*b*-PS-(20.0K). As the PEO block is chromatographically invisible, diblock C PEO(10K)-*b*-PS-(20K) and diblock B PEO(2K)-*b*-PS-(21.4K) have almost the same elution volume. Similar results were obtained with the PS-*b*-POE-*b*-PS triblock copolymers. As it seemed that the elution volume was only dependent on the total molar mass of PS in the copolymer, diblock and triblock copolymers with similar PS total chain length were compared (Figure 7).

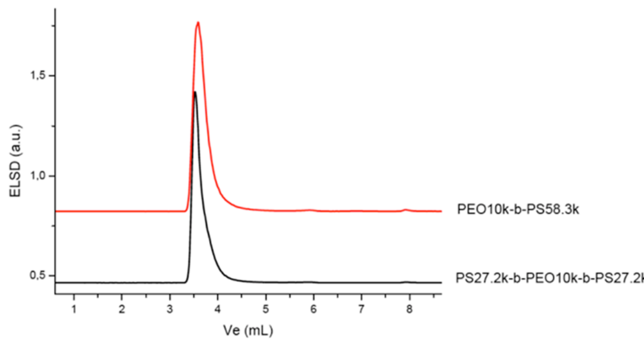

Figure 7. Chromatograms of diblock and triblock copolymers analyzed in the critical conditions of standard PEO. Stationary phase was Macherey-Nagel 250 mm × 4.6 mm Nucleosil C_8 HD, pore diameter of 100 Å, particle size of 3 μm and VWR 250 mm × 4.6 mm Nucleosil C_8, pore diameter of 120 Å, particle size of 3 μm (at 30 °C), mobile phase composition was 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v), and flow rate was 0.8 mL min^{-1}.

In the LCCC conditions, the elution volumes of the triblock PS(27.2K)-*b*-PEO(10K)-*b*-PS(27.2K) and the diblock PEO-(10K)-*b*-PS(58.3K) are rather similar, confirming the weak influence of the architecture; the retention is rather governed by the total molar mass of PS in the whole copolymer. Recently, Gorbunov and Vakhrushev[49] predicted that in the case of block copolymers containing A and B block and where the B block is under critical conditions K_{ABA} is lower than K_{AB}, K_{BAB}, or K_A. In this case, ABA block copolymer would elute earlier than BAB, AB, or A as supported by some experimental results.[50] Using Monte Carlo simulations, Wang and co-workers[51] nevertheless showed that the differences in the partition coefficients disappeared when the visible block length increased. The simulations[51] showed also that this effect tend to disappear when the size of the pore is small compared to the radius of gyration of the macromolecules.

In our case (Figure 7), the length of the visible block is higher that the invisible block, leading to a separation mainly by the whole molar mass of the visible block. For the other block copolymers where the length of the visible block is not higher than the length of the invisible block, a closer look to Figure 5 showed that the situation is slightly different. The calibration curve obtained by both the diblock and triblock copolymer followed the theory,[49,51] which means a very small over-estimation of the molar mass for triblock and a very small underestimation of the molar mass for diblock. Nevertheless, because of the use of small pore size columns, the difference of the three calibration curves is very small, and it is not possible to clearly discriminate the various architectures.

A peak is also observed around V_e = 6 mL. This small peak can be attributed to residual CH_3−PEO−acrylate or acrylate−PEO−acrylate, as already discussed above. This analysis showed that the PEO−acrylate macromonomers did not copolymerize during the styrene polymerization and was still present in the block copolymers samples after purification.

As the average chain length of the PS block were determined using ^{1}H NMR, the validity of the calibration curve for the block copolymers established using PS homopolymers was checked. The elution volume of the block copolymer in function of the average PS block length was added to the calibration curve (Figure 5). As seen in this figure, there is a good agreement between the three curves, confirming that the whole PS block length could be determined easily using the calibration curve established using PS homopolymer standards.

CONCLUSION

In this study, we developed original critical conditions with a mixture of organic solvents using reverse phase columns (C_8) for the separation of PEO. The solvent is a mixture of 58.05% chloroform, 6.45% methanol, and 35.50% *n*-heptane (v/v/v). The size of the PEO has to remain below roughly 10 000 g mol^{-1}, but the range of M_n should nevertheless be extended using larger pore size column packing material. Whereas reverse phase columns are used, the elution mechanism is governed by the interaction of polar end-group with the residual free silanol groups that are still present on the stationary phase.

In these conditions, separation of various functionalized PEO as well as the separation of diblock PEO-*b*-PS and triblock PS-*b*-PEO-*b*-PS amphiphilic copolymers were obtained. PS homopolymers and block copolymers PEO-*b*-PS and PS-*b*-PEO-*b*-PS are eluted in a SEC mode. We showed that the calibration curve obtained using PS homopolymer standards could be used to determine the whole molar mass of the PS block in block copolymers with PS and PEO moieties. In such conditions, the PEO block is invisible, and no real change according to the architecture was observed since the obtained calibration curves for di- and triblock copolymers are hardly distinguishable.

Finally, using this method, the efficiency of the synthesis of such block copolymers via the combination of 1,2 radical intermolecular addition followed by the NMP of styrene was investigated. The first step that consists in the esterification of either the HO−PEO−Me or HO−PEO−OH with an acryloyl chloride is very efficient since no residual PEO was detected. The second step that consists in the 1,2 intermolecular addition of the BlocBuilder onto the acrylate functionality to yield PEO-based macroalkoxyamines is less efficient since a small peak of residual PEO−acrylate was observed. Finally, the analysis of block copolymer revealed traces of unreacted PEO−acrylate even after purification. This method is then very efficient to monitor the preparation of block copolymer having PEO and PS blocks.

ASSOCIATED CONTENT

Supporting Information

^{1}H NMR analyses of acrylate−PEO−acrylate and BB−PEO−BB; analyses of block copolymers in PS LCCC conditions. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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